MIRZA MUHAMMAD JAMAL NASIR

RESEARCH PROFILE | ARTIFICIAL INTELLIGENCE, MACHINE LEARNING & LARGE LANGUAGE MODELS

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AI and Machine Learning Engineer with an MSc in Data Science with Advanced Research (Distinction) and 4+ years of industry experience in Machine Learning, Deep Learning, Computer Vision, NLP and MLOps. Experienced in LLM fine-tuning and evaluation using QLoRA and IA³ under the PEFT framework. Seeking a fully funded PhD focused on efficient and explainable AI, large language models and healthcare applications.

RESEARCH INTERESTS

Efficient Large Language Models (LLMs): Parameter-efficient fine-tuning using QLoRA, LoRA, IA³ and PEFT.

LLM Evaluation & Benchmarking: Model performance evaluation, comparative analysis and resource-efficient adaptation.

Trustworthy & Explainable AI: Developing interpretable, reliable and transparent machine learning systems.

AI for Healthcare: AI-driven clinical decision support and intelligent healthcare applications.

Computer Vision & Medical Imaging: Image analysis, disease detection and deep learning-based classification.

Natural Language Processing (NLP): Transformer-based models, text processing and conversational AI.

Time-Series Forecasting: Predictive modelling and data-driven forecasting.

Robust Machine Learning & MLOps: Model reliability, deployment, monitoring and scalable AI systems.

RESEARCH EDUCATION

MSc Data Science with Advanced Research (Distinction)
University of Hertfordshire, UK | September 2023 – September 2025

Dissertation: Parameter-Efficient Fine-Tuning and Evaluation of the Westlake 7.2B Language Model

Compared QLoRA and IA³ fine-tuning approaches within the Parameter-Efficient Fine-Tuning (PEFT) framework.Evaluated fine-tuned models using emotional-intelligence benchmarks.Focused on efficient adaptation of large language models under constrained computational resources.

Bachelor of Computer Software Engineering
Foundation University, Islamabad, Pakistan | September 2015 – September 2019

Final-Year Project: Event Halls Recommendation System

Combined collaborative filtering and content-based filtering techniques to develop a recommendation system.

SELECTED RESEARCH & APPLIED AI PROJECTS

LLM -Tuning & Evaluation

Compared QLoRA and IA³ parameter-efficient fine-tuning approaches for the Westlake 7.2B language model.Evaluated model performance on emotional-intelligence benchmarks as part of the MSc dissertation.

Healthcare AI & Medical Imaging

Contributed to clinical-record classification and diagnostic-data preparation in collaboration with medical professionals.Worked on skin-image classification and neural-cell detection using Mask R-CNN.

Agricultural Computer Vision

Worked on cotton-leaf disease classification and an IoT/Azure-based agricultural monitoring concept.Applied YOLOv8 to crop-anomaly detection and monitoring of crop-health and soil-moisture indicators.

Time-Series Forecasting & Predictive Modelling

Developed and evaluated forecasting approaches for financial and real-estate datasets using LSTM, ARIMA, Prophet and related methods.Explored market and social-media signals in a multimodal prediction concept.

Natural Language Processing & Conversational AI

Built a package-information chatbot using NLP techniques and NLTK.Worked with text-based prediction systems and transformer architectures.

IoT Security & Computer Vision

Worked on IoT botnet and DDoS classification using data engineering and feature-selection techniques.Applied computer-vision methods to image filtering for live news content.

RESEARCH & INDUSTRY EXPERIENCE

Machine Learning Team Lead Engineer | Kyaas Solutions | June 2023 – September 2023

Designed AI solution architectures and supervised a six-person engineering team.Coordinated MLOps integration across multiple projects, contributing to technical planning and delivery.Worked on client requirements, system architecture and project coordination.

Senior Machine Learning Engineer | Sigma Square | February 2023 – May 2023

Implemented MLflow workflows and supported cloud-based AI deployments using AWS and Microsoft Azure.

Machine Learning Engineer | University of South Asia, Centre of Innovation & Entrepreneurship | March 2022 – January 2023

Contributed to healthcare AI development, collaborating with medical professionals on dataset collection, preprocessing and annotation.

Machine Learning Engineer | Zmedia Technology | July 2018 – February 2022

Worked on machine learning and deep learning projects involving computer vision, NLP, time-series forecasting and data pipelines.

Requirement Engineer | Zmedia Technology | January 2018 – June 2018

Gathered client requirements and supported technical planning and sprint coordination..

TEACHING & MENTORING EXPERIENCE

AI and Data Science Instructor | The University of Punjab, Lahore | May 2023 – August 2023

Mentored approximately 60–70 BS, MS and PhD-level learners across multiple disciplines in research methods, Python, Machine Learning and practical AI applications.

Senior Lecturer of Computer Science | Punjab Group of Colleges | September 2022 – May 2023

Taught programming, databases, operating systems and IT fundamentals, delivering practical sessions and supporting student projects.

Lecturer of Computer Science | Noor Wings Cadet College | August 2019 – March 2020

Taught C, C++, databases and IT foundations, supporting students in core programming and computing concepts.

TECHNICAL SKILLS & TOOLS

- **Programming:** Python, C, C++, JavaScript, SQL
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras, H2O.ai, MLflow, Optuna
- **Data Science & Analytics:** NumPy, Pandas, Matplotlib, Seaborn, statistical modelling, data preprocessing and feature engineering
- **Computer Vision & NLP:** YOLO, Mask R-CNN, GANs, NLTK, spaCy, transformer models, QLoRA, LoRA, IA^3 and PEFT
- **Cloud & Development Platforms:** Microsoft Azure, AWS, Django, React.js, MERN stack, Tableau and Power BI

TRAINING, CERTIFICATIONS & RECOGNITION

- **Microsoft AI-900:** Azure Fundamentals
- **Huawei HCIA-AI:** Big Data V2.0
- **Associate Artificial Intelligence:** Machine Learning and Deep Learning — University of Punjab, Lahore (NAVTTC)
- **MERN Stack Development Bootcamp:** PSEB & P@SHA (Softoo)